14/11/2025

Ex

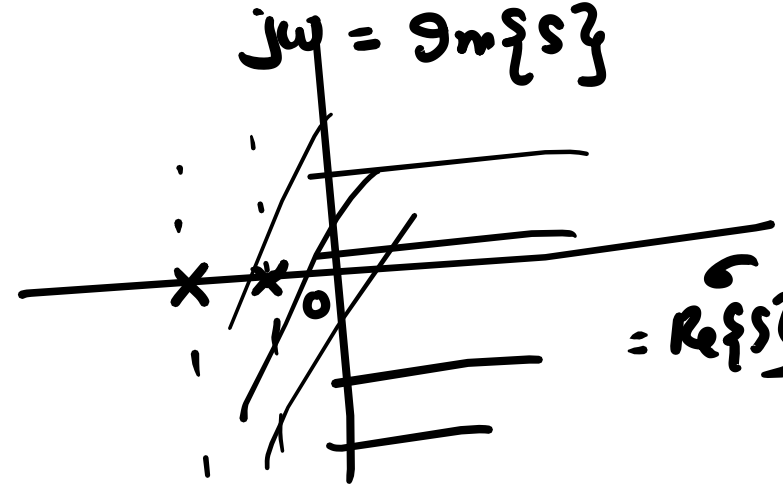
$$x(t) = 3e^{-2t}u(t) - 2e^{-t}u(t)$$

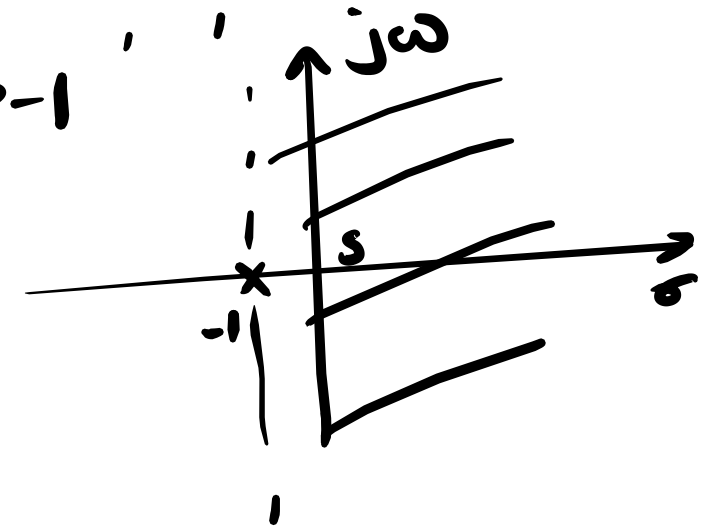
$s = \sigma + j\omega$

$$X(s) = \boxed{\frac{3}{s+2}} - \frac{2}{s+1}$$

$\Downarrow$ $\mathrm{Re}\{s\} > -2$

$\Downarrow$ $\mathrm{Re}\{s\} > -1$





$$X(s) = \frac{s-1}{(s+1)(s+2)}$$

ROC : $\mathrm{Re}\{s\} > -1$

Ex

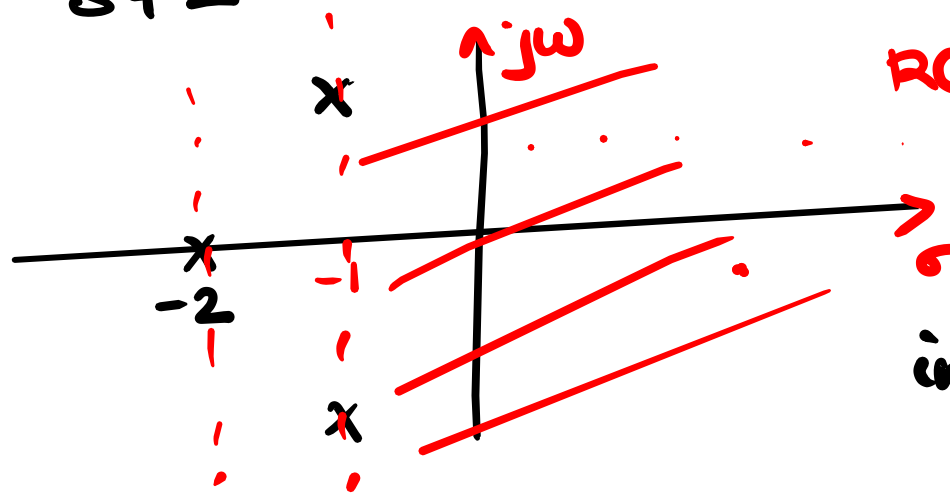
$$x(t) = e^{-2t}u(t) + e^{-t}\cos 3t\, u(t)$$

$$= e^{-2t}u(t) + e^{-t}\left(\frac{e^{j3t}+e^{-j3t}}{2}\right)u(t)$$

$$x(t) = e^{-2t}u(t) + \frac{1}{2}e^{-(1-3j)t}u(t) + \frac{1}{2}e^{-(1+3j)t}u(t)$$

$$X(s) = \frac{1}{s+2} + \frac{1}{2}\frac{1}{s+(1-3j)} + \frac{1}{2}\frac{1}{s+(1+3j)}$$

F.T. $\{x(t)e^{-\sigma t}\}$ = L.T.



ROC: $Re\{s\} > -1$

$X(s)$ is a Rational fn. in $s \equiv \frac{N(s)}{D(s)}$

No. of poles ≡ No. of zeros.

Rational fn. in s = $\frac{\text{Polynomial in } s}{\text{polynomial in } s}$ = $\frac{N(s)}{D(s)}$

$x(t) = e^{-at} u(t)$

$\Rightarrow \boxed{X(s) = \frac{1}{s+a}}$ $\quad \text{Re}\{s\} > -a$

pole ≡ where $X(s) \to \infty$

zero ≡ where $X(s) \to$ zero.

as $s \to -a$ $\quad X(s) \to \infty$

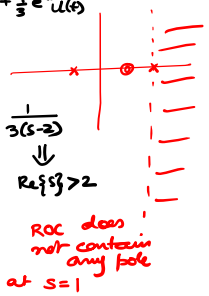
as $s \to \infty$ $\quad X(s) \to 0$

there is a pole at $s = -a$

& a zero at $s = \infty$

Ex $x(t) = \delta(t) - \frac{4}{3}e^{-t}u(t) + \frac{1}{3}e^{2t}u(t)$

$$X(s) = \frac{(s-1)^2}{(s+1)(s-2)}$$



$$= 1 - \frac{4}{3(s+1)} + \frac{1}{3(s-2)}$$

$\int_{-\infty}^{\infty} |x(t)|dt \not< \infty$

ROC ≡ entire s-plane ; $Re\{s\} > -1$; $Re\{s\} > 2$

ROC ≡ $Re\{s\} > 2$ — ROC does not contain any pole

It has a second order zero at $s = 1$

Ex $x(t) = \delta(t) - \frac{4}{3}e^{-t}u(t) + \frac{1}{3}e^{2t}u(t)$

$$X(s) = 1 - \frac{4}{3(s+1)} + \frac{1}{3(s-2)}$$

$$= \frac{3(s+1)(s-2) - 4(s-2) + (s+1)}{3(s+1)(s-2)}$$

$$= \frac{3s^2 - 3s - 6 - 4s + 8 + s + 1}{3(s+1)(s-2)}$$

$$= \frac{3s^2 - 6s + 3}{3(s+1)(s-2)} = \frac{(s-1)^2}{(s+1)(s-2)}$$